# Type MS · Bodily rhythms: circadian and ultradian

## Answer 1 of 6 · Question 32 part whole · 4 marks

2025 JUN · Paper 02 · extract 1 of 2 · find part whole in this mark scheme extract

| Question Number | Answer | Mark |
|---|---|---|
| **32** | **One** mark for identification of each way (maximum **two** marks). **One** mark for justification, which must be linked to the way identified (maximum **two** marks).<br><br>For example:<br><br>• Deandre struggles with tiredness as he is working during the circadian trough which has the lowest core body temperature and lowest production of cortisol (1), therefore during his first night shift he is working during a time when bodily functions such as alertness are low which would allow us to sleep more easily (1).<br>• The change in Deandre's exposure to the light and dark on his first night shift results in disruption to circadian rhythms as it is no longer in line with exogenous zeitgebers (1). This leads to Deandre struggling to sleep as light is a cue for the sleep-wake cycle meaning he is trying to sleep when the light is telling him he should be awake. (1).<br>• Deandre is tired due to him going to work in the dark which results in an increased production of melatonin (1). This leads to Deandre feeling tired as his body is preparing him for sleep by decreasing his alertness and temperature (1).<br><br>Accept any other appropriate response. | **(4)** |

## Type MS · Bodily rhythms: circadian and ultradian

### Answer 1 of 6 · Question 32 part whole · 4 marks

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## Type MS · Bodily rhythms: circadian and ultradian

### Answer 2 of 6 · Question 30 part whole · 1 mark

2019 JUN · Paper 02 · find part whole in this mark scheme extract

| Question number | Answer | Mark |
|---|---|---|
| **30** | **One** mark for knowledge of ultradian rhythm.<br><br>• An endogenous cycle that repeats itself in periods of less than 24 hours (1)<br><br>Accept any other appropriate response. | **(1)** |

# Type MS · Bodily rhythms: circadian and ultradian

## Answer 3 of 6 · Question 33 part whole · 9 marks

2020 NOV · Paper 02 · extract 1 of 3 · find part whole in this mark scheme extract

| Question number | Indicative content | Mark |
|---|---|---|
| **33** | AO1 = 3 marks; AO2 = 3 marks; AO3 = 3 marks<br><br>Markers must apply the descriptors in line with the general marking guidance (on pages 2–3).<br><br>Due to the intrinsic links between the skills, if a response evidences performance against only one trait/skill it will be for demonstrating knowledge and understanding. An answer displaying qualities of AO1 only, cannot be awarded more than the top of Level 1, no matter how strong the performance is in AO1.<br><br>AO1<br>• A circadian rhythm is an internal cycle in the biological processes of people that lasts approximately 24 hours.<br>• Circadian rhythms include the sleep-wake cycle that can make people sleepy and wake them up after sleeping.<br>• They are mainly endogenous but they can be affected by external cues such as light.<br><br>AO2<br>• Libby felt tired when she arrived at her destination because her sleep-wake cycle from home will be telling her to go to sleep and not stay awake for breakfast.<br>• Libby found it difficult to eat at breakfast because her body clock is still telling her it is evening, and time to go to sleep and not eat food.<br>• The external cue of daylight could help Libby adjust to her new time zone and therefore to her sleep-wake cycle, as light is a cue for her to stay awake.<br><br>AO3<br>• The circadian rhythm only focuses on biological mechanisms but individuals can override these with stimulant drugs like caffeine, so it does not fully explain her sleep pattern.<br>• Eating food can help bodily rhythms adjust to time zone differences, but one meal is not sufficient to keep Libby awake as eating does not completely 'reset' circadian rhythms.<br>• Siffre (1975) found external zeitgebers of light altered his circadian rhythm so external cues can be used to 'reset' Libby's internal cycle but this takes longer than one day. | **(9)** |

## Type MS · Bodily rhythms: circadian and ultradian

### Answer 3 of 6 · Question 33 part whole · 9 marks

2020 NOV · Paper 02 · extract 2 of 3 · find part whole in this mark scheme extract

| Level | Mark | Descriptor |
|---|---|---|
| | 0 | No rewardable material. |
| Level 1 | 1–3 | • Demonstrates isolated elements of knowledge and understanding of a limited range of psychological ideas (AO1).<br>• Provides little or no reference to relevant psychological ideas related to the context (AO2).<br>• Limited attempt to deconstruct relevant psychological ideas. Limited consideration of supporting/refuting evidence, leading to generic judgements (AO3). |
| Level 2 | 4–6 | • Demonstrates mostly accurate understanding of some relevant psychological ideas (AO1).<br>• Provides some reference to relevant psychological ideas related to the context but this may be limited or lack relevance at times (AO2).<br>• Deconstructs relevant psychological ideas using mostly logical chains of reasoning. Some consideration of supporting/refuting evidence, leading to a judgement (AO3). |
| Level 3 | 7–9 | • Demonstrates accurate and thorough knowledge and understanding of relevant psychological ideas (AO1).<br>• Provides sustained reference to relevant psychological ideas related to the context (AO2).<br>• Deconstructs relevant psychological ideas using logical chains of reasoning. Sustained consideration of supporting/refuting evidence, showing an awareness of competing arguments, leading to a judgement (AO3). |

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Answer 3 of 6 · Question 33 part whole · 9 marks

2020 NOV · Paper 02 · extract 3 of 3 · find part whole in this mark scheme extract

**Section F**
**Language, thought and communication – How do you communicate with others?**

## Type MS · Bodily rhythms: circadian and ultradian

### Answer 4 of 6 · Question 28 part whole · 1 mark

2023 JUN · Paper 02 · find part whole in this mark scheme extract

| Question number | Answer | Mark |
|---|---|---|
| **28** | **One** mark for knowledge of bodily rhythm<br><br>• A bodily rhythm is a natural, biological cycle in the body that regulates bodily functions such as sleeping (1).<br><br>Accept any other appropriate response. | **(1)** |

## Type MS · Bodily rhythms: circadian and ultradian

Answer 5 of 6 · Question 28 part whole · 1 mark

2024 JUN · Paper 02 · find part whole in this mark scheme extract

| Question number | Answer | Mark |
|---|---|---|
| **28** | D - Circadian<br>**The only correct answer is D**<br>A is incorrect as this cycle is more often than 24 hours<br>B is incorrect as this cycle is less often than 24 hours<br>C is incorrect as this cycle is annual | **(1)** |

## Type MS · Bodily rhythms: circadian and ultradian

### Answer 6 of 6 · Question 29 part whole · 2 marks

2025 JUN · Paper 02 · extract 1 of 2 · find part whole in this mark scheme extract

| Question Number | Answer | Mark |
|---|---|---|
| **29** | Up to **two** marks for understanding how the sleep cycle can be classed as an ultradian rhythm.<br><br>For example:<br><br>• Ultradian rhythms are biological cycles which repeat within 24 hours (1). The whole sleep cycle lasts around 90 mins per cycle and consists of passing through both REM and NREM sleep four or five times a night (1).<br>• Ultradian rhythms are biological cycles which last less than 24 hours (1). The whole sleep cycle repeats every 90 minutes and can be experienced 4 to 5 times a night (1).<br><br>Accept any other appropriate response. | **(2)** |

## Type MS · Bodily rhythms: circadian and ultradian

### Answer 6 of 6 · Question 29 part whole · 2 marks

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